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RESEARCH ARTICLE

Productivity Losses From Substance Use Disorder in the U.S. in 2023

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Introduction: Information on morbidity-related productivity losses attributable to substance use disorder is limited. This study estimates morbidity-related productivity losses attributable to substance use disorder among U.S. adults aged ≥18 years in 2023.

Methods: Morbidity-related productivity losses attributable to substance use disorder were estimated using the human capital approach. Using a regression-based approach, the 2021−2023 National Survey on Drug Use and Health data, along with other data and published literature, were used to estimate the cost of absenteeism, presenteeism, household productivity loss, and inability to work attributable to substance use disorder. All costs were estimated for 2023, and all analyses were conducted in 2025.

Results: Total morbidity-related productivity loss attributable to substance use disorder among U.S. adults was estimated to be $92.65 billion (95% prediction interval=$57.90, $136.48 billion) in 2023. Inability to work cost accounted for $45.25 billion (95% prediction interval=$28.29, $65.88 billion), followed by absenteeism cost of $25.65 billion (95% prediction interval=$15.11, $39.93 billion), presenteeism cost of $12.06 billion (95% prediction interval=$9.26, $15.41 billion), and cost of household productivity loss of $9.68 billion (95% prediction interval=$5.25, $15.26 billion).

Conclusions: Total morbidity-related productivity losses attributable to substance use disorder in the U.S. are substantial, amounting to $92.65 billion in 2023. Given that these estimates depend on the prevalence of substance use disorder and the amount of lost productive time, evidence-based prevention efforts and policies addressing them can help reduce these losses.
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INTRODUCTION

Morbidity-related productivity losses attributable to substance use disorder (SUD) is an important component of the economic burden of SUD, and a comprehensive understanding of this burden can provide valuable evidence to guide prevention efforts and inform decision making. SUD can impair cognitive and behavioral functioning,[1,2] resulting in productivity losses. Although previous studies have comprehensively estimated this burden for individual substances, such as cigarette smoking[3] and excessive alcohol use[4] in the U.S., and for multiple substances in Canada,[5] comprehensive estimates of morbidity-related productivity losses attributable to SUD are limited in the U.S. Existing estimates are dated[6] or underreport morbidity-related productivity losses[7−9] because they fail to consider cost of presenteeism or household productivity loss or both attributable to SUD. The U.S. Department of Justice estimated that illegal drug use resulted in a productivity loss of 17% for males and 18% for females in 2007 in the U.S.[6] Previous studies[10−12] have relied on these reductions in productivity[6] to estimate morbidity-related productivity losses attributable to prescription opioid overdose, abuse, and dependence or opioid use disorder. In addition, previous studies have used

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administrative claims data to estimate the cost of lost workplace productivity because of absenteeism or inability to work or both, attributable to prescription opioid abuse, dependence, and misuse.[7−9] However, these studies often exclude certain population groups, such as self-employed, because of convenience sampling of large employers that contribute to the administrative claims data.[7−9]

This study addresses this information gap by estimating the 2023 morbidity-related productivity losses (cost of absenteeism, presenteeism, household productivity loss, and inability to work), in total and per-affected adult, attributable to SUD among adults aged ≥18 years in the U.S.

METHODS

Study Sample

Data were obtained primarily from the National Survey on Drug Use and Health (NSDUH).[13] The NSDUH is an annual, nationally representative, cross-sectional household survey of non-institutionalized U.S. civilian population aged ≥12 years, and the survey gathers nationally representative data on the use of tobacco, alcohol, and drugs; SUDs; mental health issues; and receipt of substance use and mental health treatment from the survey respondents.[13] Adults aged ≥18 years were analyzed, excluding pregnant females because of their potentially distinct physiological and labor force participation changes affecting productivity.[14,15] Pooled 2021−2023 NSDUH data were used to obtain an adequate sample, and all analyses were conducted in 2025.

Measures

Consistent with the literature,[16] SUD was measured using a NSDUH recoded variable, which includes drug use disorder including prescription drug use disorder, among individuals who used prescription drugs in the past year.[13] Drug use disorder was defined according to DSM-5 SUD criteria for the following drugs used in the past year: marijuana, cocaine, heroin, hallucinogens, inhalants, methamphetamine, or prescription psychotherapeutic drugs (i.e., stimulants, tranquilizers or sedatives, and pain relievers).[17] Respondents with reported past-year use who met ≥2 of DSM-5 SUD criteria were classified as having a drug use disorder, hereafter referred to as SUD,[13,17] consistent with a clinically validated diagnostic framework[1,18] (Appendix Table 1, available online).

Morbidity-related productivity losses encompass cost because of absenteeism, presenteeism, household productivity loss, and inability to work. Absenteeism costs arise when employees miss workdays because of illness or injury. Presenteeism costs arise when employees are present at work but are less productive. Household productivity losses occur when individuals become unable to perform household tasks, such as cooking or cleaning. Inability to work costs arise when individuals lose productivity because of disability. Absenteeism was measured using the number of missed workdays because of illness or injury. Household productivity loss was measured using sick-days, and inability to work was measured using disability status. Considering that NSDUH does not collect information related to presenteeism, a literature-based estimate was used in estimating presenteeism days attributable to SUD (Appendix Tables 1 and 2, available online).

Statistical Analysis

This study uses human capital approach that estimates morbidity costs as the value of lost productive time resulting from illness or disability for those who had an SUD compared with those without SUD.[19] All costs were estimated by age group (18−25, 26−34, 35−49, 50−64, and ≥65 years) and sex (male and female) for individuals with SUD compared with those without SUD. To value these productivity losses, age group- and sex-specific 2016 economic (market and nonmarket) values[20] were inflation-adjusted to 2023 dollars using a gross domestic product (GDP) implicit price deflator.[21,22] A regression-based approach—accounting for survey weights and individuals' sociodemographic characteristics, health- and behavior-related variables, and year-specific effects—was used in estimating days loss because of absenteeism, home productivity loss, and inability to work, allowing for the estimation of model coefficients attributable to SUD.[3,23]

Sociodemographic variables included in regression analyses were age (18−25, 26−34, 35−49, 50−64, or ≥65 years), sex (male or female), marital status (married, widowed, divorced or separated, or never married), race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, or non-Hispanic other [non-Hispanic Asian or non-Hispanic all other races]), educational attainment (less than high school, high school diploma or GED, some college, associate degree, or college or above), annual family income (income below federal poverty level [FPL], 100%−200% FPL, or >200% FPL), and urban/rural status (large metro, small metro, or nonmetro). The health- and behavior-related variables included were BMI (weight [in kg]/height [in meter]^2; categories: underweight [<18.5 kg/m^2], normal weight [18.5 to <25 kg/m^2], overweight [25 to <30 kg/m^2], or obese [≥30 kg/m^2]), health insurance coverage, alcohol use (binge alcohol use, past 30 days), and cigarette use (days of cigarette used, past 30 days).

To calculate age group- and sex-specific absenteeism cost, age group- and sex-specific absenteeism days during the past 30 days were estimated for adults who had an SUD and were employed, using a standard negative binomial model, which was selected based on specification tests for model selection and Akaike and Bayesian information criteria, with the 2021−2023 NSDUH data (Appendix Tables 2 and 3, available online).[3,23] The resulting absenteeism days were multiplied by 12 to estimate annual absenteeism days. The age group- and sex-specific annual absenteeism days (Appendix Table 5, available online) were multiplied by the corresponding number of adults with SUD (Appendix Table 4, available online) who were employed in 2023 to calculate total absenteeism days attributable to SUD in 2023. Finally, the resulting age group- and sex-specific total absenteeism days were multiplied by the corresponding daily market economic value—representing daily gross earning adjusted for employer-paid benefits[20]—to calculate absenteeism cost attributable to SUD by age group and sex. To calculate age group- and sex-specific daily market economic value, the 2016 annual market economic values[20] were inflation-adjusted to 2023 dollars using the GDP deflator[21] and divided by 250, assuming 250 workdays in a year (Appendix Tables 1 and 2, available online).[3,23]

To calculate age group- and sex-specific presenteeism cost, age group- and sex-specific annual presenteeism days were estimated by multiplying the total number of average annual workdays after accounting for absenteeism days (i.e., 250 days minus absenteeism days) by a presenteeism rate of 1.685% based on estimates from a previous study on current smokers[24] and applied to adults with SUD (Appendix Table 4, available online) who were employed. The resulting presenteeism days (Appendix Table 5, available online) were then multiplied by the corresponding number of adults with SUD who were employed in 2023 to calculate total presenteeism days attributable to SUD. Finally, age group- and sex-specific presenteeism days were multiplied by the corresponding daily market economic values—consistent with those used in absenteeism cost—to calculate presenteeism cost attributable to SUD[3,23] (Appendix Tables 1 and 2, available online).

To calculate age group- and sex-specific inability to work cost, the age groups- and sex-specific proportion of adults who were unable to work because of SUD was estimated using a multivariate logistic regression model with using the 2021−2023 NSDUH data (Appendix Tables 2 and 3, available online). These age group- and sex-specific proportions (Appendix Table 5, available online) were then multiplied by the corresponding number of adults with SUD (Appendix Table 4, available online) to calculate age group- and sex-specific total number of adults unable to work attributable to SUD. Finally, age group- and sex-specific inability to work costs were estimated by multiplying age group- and sex-specific number of adults who were unable to work because of SUD-related disability by corresponding market economic values—as used in absenteeism cost (Appendix Tables 1 and 2, available online).[3,23]

To calculate age group- and sex-specific cost of household productivity loss attributable to SUD, the age group- and sex-specific annual sick days were estimated using a negative binomial model, based on specification tests for model selection and Akaike and Bayesian information criteria, using the 2021−2023 NSDUH data (Appendix Tables 2 and 3, available online). Given that the NSDUH includes sick days resulting in hospital stays and only a small portion of sickness or injury cases necessitate hospitalization,[25] the estimated number of sick-days were adjusted using the average number of hospital-based sick-days from the NSDUH data[13] and average number of sick-days in hospital and at home from National Health Interview Survey data[26] during 2015−2018 (Appendix Tables 1 and 2, available online). The National Health Interview Survey is an annual U.S. household survey collecting nationally representative data on health, behavior, and demographic characteristics of the U.S. civilian non-institutional population.[26] The adjusted age group- and sex-specific sick-days (Appendix Table 5, available online) were then multiplied by the corresponding number of adults with SUD (Appendix Table 4, available online) to estimate total sick days attributable to SUD. The age group- and sex-specific cost of household productivity loss was estimated by multiplying the corresponding sick days by daily value of household production, which reflects value of nonmarket time spent in household caring and volunteer services.[3,23] The 2016 annual household production values were inflation-adjusted to 2023 dollars using the GDP deflator and divided by 365 days—assuming that adults engage in nonmarket production every day—to calculate daily value of household production (Appendix Tables 1 and 2, available online).

Given that absenteeism and presenteeism apply to employed adults, these costs were estimated for employed adults. The costs of inability to work and household productivity loss were estimated for all individuals.[3] Considering that sick days in hospital beds or at home would also result in absenteeism days for employed individuals, including adults with sick days in the absenteeism cost estimate does not result in double counting. However, including individuals who were unable to work in household productivity cost may lead to double counting because productivity loss from being unable to work and being hospitalized cannot occur at the same time, and the modeling approach excluded this double counting.[3]

The cost of morbidity-related productivity losses attributable to SUD was derived by summing the 4 cost components, with estimated costs reported per adult with an SUD. Stata, version 18.5 (StataCorp, LLC) was used for the analysis. Given that this analysis was based on secondary and deidentified data, an IRB review was not required.

RESULTS

In 2023, the estimated cost of morbidity-related productivity losses attributable to SUD in the U.S. was $92.65 billion (95% prediction interval [PI]=$57.90, $136.48 billion) (Table 1). Of this total cost, males accounted for $61.19 billion (95% PI=$38.86, $89.25 billion) and females accounted for $31.45 billion (95% PI=$19.05, $47.24 billion). By cost component, the inability to work cost was estimated to be $45.25 billion (95% PI=$28.29, $65.88 billion), with males accounting for $31.30 billion (95% PI=$19.67, $45.43 billion) and females accounting for $13.95 billion (95% PI=$8.62, $20.44 billion). The absenteeism cost was estimated at $25.65 billion (95% PI=$15.11, $39.93 billion), with males accounting for $17.06 billion (95% PI=$10.23, $26.24 billion) and females accounting for $8.59 billion (95% PI: $4.88, $13.69 billion). The presenteeism cost was estimated at $12.06 billion (95% PI=$9.26, $15.41 billion), with males accounting for $8.69 billion (95% PI=$6.72, $11.05 billion) and females accounting for $3.37 billion (95% PI=$2.54, $4.36 billion). Finally, the cost of household productivity loss was estimated to be $9.68 billion (95% PI=$5.25, $15.26 billion), with males contributing $4.14 billion (95% PI=$2.23, $6.52 billion) and females contributing $5.55 billion (95% PI=$3.02, $8.74 billion). In general, females experienced greater loss of productive time because of SUD. SUD-related absenteeism days were highest among adults aged 18−25 years, whereas presenteeism and sick days were highest among those aged ≥65 years. The highest proportion of adults unable to work because of SUD was observed among those aged 50−64 years (Appendix Table 5, available online). Generally, adults aged 35−49 years incurred the highest total productivity loss costs because of absenteeism, presenteeism, and household productivity loss, whereas adults aged 50−64 years experienced the highest productivity loss cost because of inability to work (Appendix Table 6, available online).

In 2023, the estimated cost of morbidity-related productivity losses per adult with SUD was $3,703 (95% PI=$2,684, $4,795) (Table 2). The estimated per adult cost was $4,242 for males (95% PI=$3,101, $5,467) and $2,969 for females (95% PI=$2,105, $3,891). By cost component, the cost of inability to work per adult with

Table 1. Productivity Losses Attributable to SUD in the U.S. in 2023

Cost component	Male	Female	Total
Absenteeism[a]	17.06	8.59	25.65
	(10.23, 26.24)	(4.88, 13.69)	(15.11, 39.93)
Presenteeism[b]	8.69	3.37	12.06
	(6.72, 11.05)	(2.54, 4.36)	(9.26, 15.41)
Household productivity loss[c]	4.14	5.55	9.68
	(2.23, 6.52)	(3.02, 8.74)	(5.25, 15.26)
Inability to work[d]	31.30	13.95	45.25
	(19.67, 45.43)	(8.62, 20.44)	(28.29, 65.88)
Total cost[e]	61.19	31.45	92.65
	(38.86, 89.25)	(19.05, 47.24)	(57.90, 136.48)

Note: All costs were estimated in the 2023 US$ (in billion). Values in parentheses are 95% prediction interval, calculated based on the 95% CI of model-based estimates. Each cost component was calculated as the sum of each cost component over all age groups.
[a]Absenteeism cost attributable to SUD was calculated by multiplying the age group- and sex-specific missed workdays because of absenteeism attributable to SUD for those who were employed, using the age group- and sex-specific market economic value from Grosse et al.,[20] which was inflation-adjusted to the 2023 US$.
[b]Presenteeism cost attributable to SUD was calculated by multiplying the age group- and sex-specific missed workdays because of presenteeism attributable to SUD for those who were employed, using the age group- and sex-specific market economic value from Grosse et al.,[20] which was inflation-adjusted to the 2023 US$.
[c]Cost of household productivity loss attributable to SUD was calculated by multiplying the age group- and sex-specific sick-days attributable to SUD for all adults, using the age group- and sex-specific nonmarket economic value from Grosse et al.,[20] which was inflation-adjusted to the 2023 US$.
[d]Cost of inability to work attributable to SUD was calculated by multiplying the age group- and sex-specific fraction of total adults who had inability to work attributable to SUD, using the age group- and sex-specific market economic value from Grosse et al.,[20] which was inflation-adjusted to the 2023 US$.
[e]Total cost was calculated as the sum of the cost of absenteeism, presenteeism, household productivity loss, and inability to work.
SUD, substance use disorder.

Table 2. Productivity Losses Attributable to SUD per Adult with SUD in the U.S. in 2023

Cost component	Male	Female	Total
Absenteeism[a]	1,183	811	1,025
	(817, 1,607)	(539, 1,128)	(700, 1,402)
Presenteeism[b]	603	318	482
	(536, 677)	(281, 359)	(429, 541)
Household productivity loss[c]	287	524	387
	(178, 400)	(333, 720)	(243, 536)
Inability to work[d]	2,170	1,316	1,808
	(1,570, 2,783)	(952, 1,684)	(1,311, 2,315)
Total per adult cost[e]	4,242	2,969	3,703
	(3,101, 5,467)	(2,105, 3,891)	(2,684, 4,795)

Note: All costs (in the 2023 US$) were estimated per adult aged ≥18 years who had SUD. Values in parentheses are 95% prediction interval, calculated based on the 95% CI of model-based estimates. Each cost component was calculated as the sum of each cost component over all age groups.
[a]Per person absenteeism cost attributable to SUD was calculated by dividing total absenteeism cost attributable to SUD by the number of adults aged ≥18 years who had an SUD in 2023.
[b]Per person presenteeism cost attributable to SUD was calculated by dividing total presenteeism cost attributable to SUD by the number of adults aged ≥18 years who had an SUD in 2023.
[c]Per person cost of household productivity loss attributable to SUD was calculated by dividing the total cost of household productivity loss attributable to SUD by the number of adults aged ≥18 years who had an SUD in 2023.
[d]Per person cost of inability to work attributable to SUD was calculated by dividing the total cost of inability to work attributable to SUD by the number of adults aged ≥18 years who had an SUD in 2023.
[e]Total per adult cost was calculated as the sum of the per adult cost of absenteeism, presenteeism, household productivity loss, and inability to work.
SUD, substance use disorder.

SUD was estimated at $1,808 (95% PI=$1,311, $2,314), with this cost being $2,170 (95% PI=$1,570, $2,783) for males and $1,316 (95% PI=$952, $1,684) for females. The cost of absenteeism per adult with SUD was estimated to be $1,025 (95% PI=$700, $1,403), with cost per adult of $1,183 (95% PI=$817, $1,607) for males and $811 (95% PI=$539, $1,128) for females. The presenteeism cost per adult with SUD was estimated at $482 (95% PI=$429, $541), with this cost being $603 (95% PI=$536, $677) for males and $318 (95% PI=$281, $359) for females. Finally, the cost of household productivity loss per adult with SUD was estimated at $387 (95% PI= $243, $536), with $287 (95% PI=$178, $400) for males and $524 (95% PI=$333, $720) for females. The per adult cost of absenteeism and presenteeism were higher among adults aged 35−49 years, whereas the per adult cost of inability to work and household productivity loss were higher among adults aged 50−64 and ≥65 years (Appendix Table 7, available online).

DISCUSSION

This study is the first of its kind to provide comprehensive cost estimates of the morbidity-related productivity losses attributable to SUD in the U.S. The estimated productivity losses of $92.65 billion or $3,703 per adult with an SUD in 2023 are substantial. Despite a greater loss of productive time because of SUD among females compared with that among males, the overall cost, along with 3 cost components—absenteeism, presenteeism, and inability to work—was larger for males primarily because of higher market economic values[20] to males. The cost of household productivity loss was larger for females because of higher nonmarket economic values[20] to females. Total costs because of absenteeism, presenteeism, and household productivity loss were generally higher among adults aged 35−49 years not necessarily because of greater days lost but result from higher economic values.[20] The wide prediction intervals for many of the reported results are because of the use of regression results. Consistent with previous studies,[3,23] inability to work accounted for the largest share (49%) of the total cost. Simultaneously, absenteeism costs accounted for 28% of the total costs, followed by presenteeism costs (13%) and the cost of household productivity loss (10%).

The estimated cost in this study is larger than those reported in previous studies[7−11] for certain reasons. This study focused on drug use disorder, whereas previous studies focused on opioid use disorder[11] or prescription opioid overdose, abuse, and dependence,[7] which have lower prevalence rates than this study's SUD indicator. In addition, the findings of the studies[7−9] using administrative claims data may not be generalized at the population level given the convenience sample of large employers contributing to the data.

Given that these cost estimates depend on the amount of lost productive time and SUD prevalence, implementing evidence-based prevention strategies aimed at improving access to SUD treatment and reducing harm through the provision of medications and therapeutic interventions may help reduce the loss of productive time.[27] Regulating the supply and legal access to approved medications, along with educating patients about opioids and other prescribed drugs, may help reduce the prevalence of SUD.[27] Given that SUD-attributable absenteeism days were highest among adults aged 18−25 years, youth-focused strategies —such as strengthening family support; enhancing parenting practices[28,29] that support healthy development and drug refusal and social skills[30]; creating protective community environments that limit exposure to substance use and promote healthly social norms[31,32]; engaging youths in structured activities that reduce unsupervised time[33,34]; and providing support to those at increased risk of substance use such as individuals with adverse childhood experiences[35]—may help prevent youth substance use and reduce related productivity losses. Effective prevention and treatment require integrated, ethically grounded approaches that address both individual risk factors and broader social determinants through coordinated efforts across health, social, and justice systems.[1,36]

Limitations

Limitations include the reliance on self-reported SUD responses and the exclusion of nonhoused populations, such as homeless or incarcerated individuals, that have higher SUD prevalence rates.[37] For example, the prevalence of opioid use disorder in Massachusetts in 2015 was 4.49 times higher than NSDUH-based estimates, suggesting that NSDUH-based productivity losses may be similarly underestimated if the Massachusetts rate reflects the national estimates.[38] Although some previous studies have attempted to correct for prevalence underreporting, most rely on localized data[38,39] or use noncomparable definitions.[40−42] This study uses the 2021−2023 NSDUH data, which reflect DSM-5 criteria and partially address underreporting.[43] To maintain methodological consistency,[3,23] unadjusted prevalence estimates were used, which may result in conservative cost estimates. The use of human capital approach may overestimate productivity losses compared with friction cost approach.[19] In addition, human capital approach may not accurately reflect the value of time because it incorporates differences in earnings by sex. Moreover, using the respondents' disability as the primary reason for not working in the past week as a measure for inability to work may lead to an overestimation of the inability to work cost. Furthermore, the estimated cost of productivity losses does not account for the caregiving costs and the impact on family members who experienced productivity losses as a result of their loved one's SUD. The estimates of presenteeism cost rely on presenteeism rates associated with current smoking because of the lack of nationally representative and reliable SUD-related presenteeism rates. Although smoking-related presenteeism may result from mild withdrawal and reduced focus,[24,44] SUDs may involve more severe cognitive, psychiatric, and physiological impairments,[45,46] likely causing greater productivity losses. Therefore, using smoking-related presenteeism rates may underestimate SUD-related presenteeism costs. Furthermore, substance-specific estimates were not reported because of limited sample sizes within subgroups stratified by age group and sex. Finally, the estimated productivity losses do not account for productivity losses among adults who did not have an SUD last year but had an SUD in previous years. Future studies should address these issues to improve the accuracy and scope of SUD-related productivity losses to better guide targeted policy and prevention efforts.

CONCLUSIONS

In 2023, the morbidity-related productivity losses from SUD among U.S. adults were substantial, reaching $92.65 billion. Public health strategies addressing the prevalence of SUD and reducing lost productive time have the potential to reduce these losses and offer cost savings for the U.S. economy.

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CREDIT AUTHOR STATEMENT

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SUPPLEMENTAL MATERIAL

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REFERENCES

1. Volkow ND, Blanco C. Substance use disorders: a comprehensive update of classification, epidemiology, neurobiology, clinical aspects, treatment and prevention. *World Psychiatry*. 2023;22(2):203–229. https://doi.org/10.1002/wps.21073.
2. McLellan AT, Lewis DC, O'brien CP, Kleber HD. Drug dependence, a chronic medical illness: implications for treatment, insurance, and outcomes evaluation. *JAMA*. 2000;284(13):1689–1695. https://doi.org/10.1001/jama.284.13.1689.
3. Shrestha SS, Ghimire R, Wang X, Trivers KF, Homa DM, Armour BS. Cost of cigarette smoking–attributable productivity losses, U.S., 2018. *Am J Prev Med*. 2022;63(4):478–485. https://doi.org/10.1016/j.amepre.2022.04.032.
4. Sacks JJ, Gonzales KR, Bouchery EE, Tomedi LE, Brewer RD. 2010 national and state costs of excessive alcohol consumption. *Am J Prev Med*. 2015;49(5):e73–e79. https://doi.org/10.1016/j.amepre.2015.05.031.
5. Canadian Substance Use Costs and Harms Scientific Working Group, Canadian Substance Use Costs and Harms (2007−2014), 2018, Canadian Centre on Substance Use and Addiction; Ottawa. https://www.ccsa.ca/sites/default/files/2019-04/CSUCH-Canadian-Substance-Use-Costs-Harms-Report-2018-en.pdf. Accessed June 20, 2025.
6. National Drug Intelligence Center, The Economic Impact of Illicit Drug Use on American Society, 2011, United States Department of Justice; Washington, D.C. https://www.justice.gov/archive/ndic/pubs44/44731/44731p.pdf. Accessed June 22, 2025.
7. Birnbaum HG, White AG, Schiller M, Waldman T, Cleveland JM, Roland CL. Societal costs of prescription opioid abuse, dependence, and misuse in the United States. *Pain Med*. 2011;12(4):657–667. https://doi.org/10.1111/j.1526-4637.2011.01075.x.
8. Rice JB, Kirson NY, Shei A, et al. Estimating the costs of opioid abuse and dependence from an employer perspective: a retrospective analysis using administrative claims data. *Appl Health Econ Health Policy*. 2014;12(4):435–446. https://doi.org/10.1007/s40258-014-0102-0.
9. Henke RM, Ellsworth D, Wier L, Snowdon J. Opioid use disorder and employee work presenteeism, absences, and health care costs. *J Occup Environ Med*. 2020;62(5):344–349. https://doi.org/10.1097/JOM.0000000000001830.
10. Florence CS, Zhou C, Luo F, Xu L. The economic burden of prescription opioid overdose, abuse, and dependence in the United States, 2013. *Med Care*. 2016;54(10):901–906. https://doi.org/10.1097/MLR.0000000000000625.
11. Florence C, Luo F, Rice K. The economic burden of opioid use disorder and fatal opioid overdose in the United States, 2017. *Drug Alcohol Depend*. 2021;218:108350. https://doi.org/10.1016/j.drugalcdep.2020.108350.
12. Luo F, Li M, Florence C. State-level economic costs of opioid use disorder and fatal opioid overdose—United States, 2017. *MMWR Morb Mortal Wkly Rep*. 2021;70(15):541–546. https://doi.org/10.15585/mmwr.mm7015a1.
13. Substance Abuse and Mental Health Services Administration. National Survey on Drug Use and Health (NSDUH). https://www.samhsa.gov/data/data-we-collect/nsduh-national-survey-drug-use-and-health. Accessed February 19, 2025.
14. Declercq E, Sakala C, Corry M, Applebaum S, Herrlich A. Listening to Mothers^{SM}III: New Mothers Speak Out, 2013, Childbirth Connection; New York, NY. https://nationalpartnership.org/wp-content/uploads/2023/02/listening-to-mothers-iii-new-mothers-speak-out-2013.pdf. Accessed June 20, 2025.
15. Wheeler SM, Massengale KEC, Adewumi K, et al. Pregnancy vs. paycheck: a qualitative study of patient's experience with employment during pregnancy at high risk for preterm birth. *BMC Pregnancy Childbirth*. 2020;20(1):565. https://doi.org/10.1186/s12884-020-03246-7.
16. Liu L, Zhang C, Nahata MC. Trends in treatment need and receipt for substance use disorders in the U.S. *JAMA Netw Open*. 2025;8(1): e2453317. https://doi.org/10.1001/jamanetworkopen.2024.53317.
17. Substance Abuse and Mental Health Services Administration. Key Substance Use and Mental Health Indicators in the United States: Results From the 2022 National Survey on Drug Use and Health. Center for Behavioral Health Statistics and Quality, 2024, Substance Abuse and Mental Health Services Administration. 2025;22 https://www.samhsa.gov/data/sites/default/files/reports/rpt42731/2022-nsduh-nnr.pdf.
18. Hasin DS, O'brien CP, Auriacombe M, et al. DSM-5 criteria for substance use disorders: recommendations and rationale. *Am J Psychiatry*. 2013;170(8):834–851. https://doi.org/10.1176/appi.ajp.2013.12060782.
19. Pike J, Grosse SD. Friction cost estimates of productivity costs in cost-of-illness studies in comparison with human capital estimates: a review. *Appl Health Econ Health Policy*. 2018;16(6):765–778. https://doi.org/10.1007/s40258-018-0416-4.
20. Grosse SD, Krueger KV, Pike J. Estimated annual and lifetime labor productivity in the United States, 2016: implications for economic evaluations. *J Med Econ*. 2019;22(6):501–508. https://doi.org/10.1080/13696998.2018.1542520.
21. Gross domestic product: implicit price deflator (GDPDEF). Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/GDPDEF#0. Updated August 29, 2024. Accessed January 15, 2025.
22. Dunn A, Grosse SD, Zuvekas SH. Adjusting health expenditures for inflation: a review of measures for health services research in the United States. *Health Serv Res*. 2018;53(1):175–196. https://doi.org/10.1111/1475-6773.12612.
23. Shrestha SS, Honeycutt AA, Yang W, et al. Economic costs attributable to diabetes in each U.S. state. *Diabetes Care*. 2018;41(12):2526–2534. https://doi.org/10.2337/dc18-1179.
24. Bunn WB III, GM Stave, KE Downs, JMJ Alvir, R Dirani. Effect of smoking status on productivity loss. *J Occup Environ Med*. 2006;48(10):1099–1108. https://doi.org/10.1097/01.jom.0000243406.08419.74.
25. Emergency department visits. National Center for Health Statistics, Centers for Disease Control and Prevention. https://www.cdc.gov/nchs/fastats/emergency-department.htm. Updated September 6, 2022. Accessed January 4, 2025.
26. National health interview survey 1997−2018. National Center for Health Statistics, Centers for Disease Control and Prevention. https://www.cdc.gov/nchs/nhis/1997-2018.htm. Updated XXX. Accessed December 10, 2025.
27. National Academies of Sciences Engineering and Medicine, Evidence on strategies for addressing the opioid epidemic, In: J.K. Phillips, M.A. Ford and R.J. Bonnie, Pain Management and the Opioid Epidemic: Balancing Societal and Individual Benefits and Risks of Prescription Opioid Use, 2017, National Academies Press (U.S.); Washington, DC. https://www.ncbi.nlm.nih.gov/books/NBK458653/. Acessed June 25, 2024.
28. Griffin KW, Botvin GJ. Evidence-based interventions for preventing substance use disorders in adolescents. *Child Adolesc Psychiatr Clin N Am*. 2010;19(3):505–526. https://doi.org/10.1016/j.chc.2010.03.005.
29. Allen ML, Garcia-Huidobro D, Porta C, et al. Effective parenting interventions to reduce youth substance use: a systematic review. *Pediatrics*. 2016;138(2):e20154425. https://doi.org/10.1542/peds.2015-4425.
30. Nawi AM, Ismail R, Ibrahim F, et al. Risk and protective factors of drug abuse among adolescents: a systematic review. *BMC Public Health*. 2021;21(1):2088. https://doi.org/10.1186/s12889-021-11906-2.
31. Hays SP, Hays CE, Mulhall PF. Community risk and protective factors and adolescent substance use. *J Prim Prev*. 2003;24(2):125–142. https://doi.org/10.1023/A:1025940311556.
32. Brown EC, Hawkins JD, Rhew IC, et al. Prevention system mediation of communities that care effects on youth outcomes. *Prev Sci*. 2014;15(5):623–632. https://doi.org/10.1007/s11121-013-0413-7.

33. Farb AF, Matjasko JL. Recent advances in research on school-based extracurricular activities and adolescent development. *Dev Rev*. 2012;32(1):1–48. https://doi.org/10.1016/j.dr.2011.10.001.
34. Bond L, Butler H, Thomas L, et al. Social and school connectedness in early secondary school as predictors of late teenage substance use, mental health, and academic outcomes. *J Adolesc Health*. 2007;40(4):357.e9–357.e18. https://doi.org/10.1016/j.jadohealth.2006.10.013.
35. Shin SH, McDonald SE, Conley D. Patterns of adverse childhood experiences and substance use among young adults: a latent class analysis. *Addict Behav*. 2018;78:187–192. https://doi.org/10.1016/j.addbeh.2017.11.020.
36. World Health Organization, United Nations Office on Drugs and Crime. Incorporating results of field-testing, International Standards for the Treatment of Drug Use Disorders, Revised Edition, 2020, World Health Organization and United Nations Office on Drugs and Crime; Geneva. https://www.unodc.org/documents/drug-prevention-and-treatment/UNODC-WHO_International_Standards_Treatment_Drug_Use_Disorders_April_2020.pdf. Accessed June 22, 2025.
37. Reuter P, Caulkins JP, Midgette G. Heroin use cannot be measured adequately with a general population survey. *Addiction*. 2021;116(10):2600–2609. https://doi.org/10.1111/add.15458.
38. Barocas JA, White LF, Wang J, et al. Estimated prevalence of opioid use disorder in Massachusetts, 2011−2015: a capture−recapture analysis. *Am J Public Health*. 2018;108(12):1675–1681. https://doi.org/10.2105/AJPH.2018.304673.
39. Doogan NJ, Mack A, Wang J, et al. Opioid use disorder among Ohio's Medicaid population: prevalence estimates from 19 counties using a multiplier method. *Am J Epidemiol*. 2022;191(12):2098–2108. https://doi.org/10.1093/aje/kwac154.
40. Keyes KM, Rutherford C, Hamilton A, et al. What is the prevalence of and trend in opioid use disorder in the United States from 2010 to 2019? Using multiplier approaches to estimate prevalence for an unknown population size. *Drug Alcohol Depend Rep*. 2022;3:100052. https://doi.org/10.1016/j.dadr.2022.100052.
41. Lim TY, Keyes KM, Caulkins JP, Stringfellow EJ, Cerdá M, Jalali MS. Improving estimates of the prevalence of opioid use disorder in the United States: revising Keyes et al. *J Addict Med*. 2024;18(6):705–710. https://doi.org/10.1097/ADM.0000000000001375.
42. Levy NS, Palamar JJ, Mooney SJ, Cleland CM, Keyes KM. What is the prevalence of drug use in the general population? Simulating underreported and unknown use for more accurate national estimates. *Ann Epidemiol*. 2022;68:45–53. https://doi.org/10.1016/j.annepidem.2021.12.013.
43. Hasin D, Stohl M. Higher prevalence estimates of substance use disorders with DSM-5 versus DSM-IV criteria among U.S. nonelderly adults with substance use: the role of DSM-IV diagnostic orphans. *Am J Psychiatry*. 2024;181(11):955–957. https://doi.org/10.1176/appi.ajp.20240852.
44. Parrott AC. Cigarette-derived nicotine is not a medicine. *World J Biol Psychiatry*. 2003;4(2):49–55. https://doi.org/10.3109/15622970309167951.
45. Volkow ND, Koob GF, McLellan AT. Neurobiologic advances from the brain disease model of addiction. *N Engl J Med*. 2016;374(4):363–371. https://doi.org/10.1056/NEJMra1511480.
46. Frone MR. Prevalence and distribution of alcohol use and impairment in the workplace: a U.S. national survey. *J Stud Alcohol*. 2006;67(1):147–156. https://doi.org/10.15288/jsa.2006.67.147.